# 3 System response

## 3.1 Introduction

How will the output from a system change when there is an input to it? For example, if we consider a mercury-in-glass thermometer as a system with an input of temperature and an output of the level of the mercury in the glass capillary, how will the level change when the thermometer is suddenly immersed in hot water, i.e. given a step input? As a matter of experience we know that the level will increases as a result of the input and take a certain amount of time before it reaches its steady state value. If we consider a control system employing a motor and feedback to move a work piece in an automatic machining operation, how will the output, i.e. the displacement of the workpiece, vary with time when the input is gradually increased with time with the aim of gradually increasing the displacement of the workpiece? This chapter is concerned with a method we can use to answer this question and, in general, determine how the output of systems changes when there is a change in input.

In Chapter 2 the method of describing a system by means of a transfer function was introduced; the transfer function is the output divided by the input when both are written as $s$ functions. In this chapter we consider how we can use transfer functions to determine how the output of a system will change with time for particular inputs.

## 3.2 Inputs

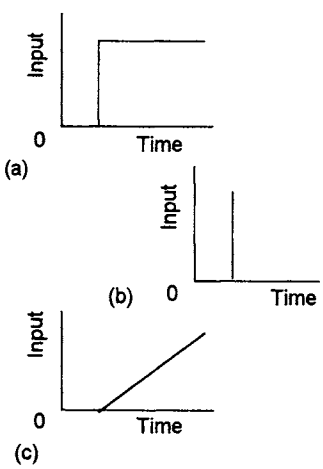


Figure 3.1 *Forms of input: (a) step, (b) impulse, (c) ramp*

Inputs to systems commonly take a number of standard forms (Figure 3.1). With the *step input* we have the input suddenly being switched to a constant value at some particular time. With the *impulse input* we have the input existing for just a very brief time before dropping back to zero. With the *ramp input*, we have the input starting at some time and then increasing at a constant rate. We can write such inputs as $s$ functions in the following way:

1. A *unit step input* which starts at a time $t = 0$ and rises to the constant value 1 has a Laplace transform of the input as an $s$ function multiplied by $1/s$.
2. A *unit impulse input* which starts at a time $t = 0$ and rises to the value 1 has a Laplace transform of the input as an $s$ function multiplied by 1.
3. A *unit ramp input* which starts at time $t = 0$ and rises by 1 each second has a Laplace transform of the input as an $s$ function multiplied by $1/s^2$.

In general, if a function of time is multiplied by some constant, then the Laplace transform of that function is multiplied by the same constant. Thus, if we have a step input of size 5 then the Laplace transform is 5 times the transform of a unit step and so the input as an $s$ function is $5/s$.

**Example**

An electrical system has an input of a voltage of 2 V which is suddenly applied by a switch being closed. What is the input as an $s$ function?

Assume the input occurs at time $t = 0$. The input is a step voltage of size 2 V. An input of a unit step voltage as an $s$ function is $(1/s)$ and thus for a 2 V step is $(2/s)$ V.

**Example**

A controlled speed motor has a voltage input which is increased at the rate of 3 V per second. What is the input as an $s$ function?

Assume the input starts at time t = 0. The input is a ramp voltage of 3 V/s. An input of a unit ramp voltage as an $s$ function is $(1/s^2)$ and thus for a 3 V/s ramp is $(3/s^2)$.

## 3.3 Determining outputs

The procedure we can use to determine how the output of a system will change with time when there is some input to the system is:

1 *Determine the output as an s function*
In terms of the transfer function $G(s)$ we have:

$$\text{Output } (s) = G(s) \times \text{Input } (s)$$

We can thus obtain the output of a system as an $s$ function by multiplying its transfer function by the input $s$ function.

2 *Determine the time function corresponding to the output s function*
To obtain the output as a function of time we need to find the time function that will give the particular output $s$ function that we have obtained. Tables of $s$ functions and their corresponding time functions can be used; Table 3.1 is a table of commonly encountered functions. Often, however, the $s$ function output has to be rearranged to put it into a form given in the table.

In obtaining the time function corresponding to a particular $s$ function we can utilise the following properties of $s$ functions:

1 If a Laplace transform is multiplied by some constant then the corresponding time function is multiplied by the same constant. For example, if we have $3/(s + 1)$ then the corresponding time function is that of 3 times the time function for $1/(s + 1)$.

Table 3.1 *Laplace functions and their corresponding time functions*

| | Time function $f(t)$ | Laplace transform $F(s)$ |
|---|---|---|
| 1 | A unit impulse | $1$ |
| 2 | A unit step | $\frac{1}{s}$ |
| 3 | $t$, a unit ramp | $\frac{1}{s^2}$ |
| 4 | $e^{-at}$, exponential decay | $\frac{1}{s+a}$ |
| 5 | $1-e^{-at}$, exponential growth | $\frac{a}{s(s+a)}$ |
| 6 | $te^{-at}$ | $\frac{1}{(s+a)^2}$ |
| 7 | $t-\frac{1-e^{-at}}{a}$ | $\frac{a}{s^2(s+a)}$ |
| 8 | $e^{-at}-e^{-bt}$ | $\frac{b-a}{(s+a)(s+b)}$ |
| 9 | $(1-at)e^{-at}$ | $\frac{s}{(s+a)^2}$ |
| 10 | $1-\frac{b}{b-a}e^{-at}+\frac{a}{b-a}e^{-bt}$ | $\frac{ab}{s(s+a)(s+b)}$ |
| 11 | $\frac{e^{-at}}{(b-a)(c-a)}+\frac{e^{-bt}}{(c-a)(a-b)}+\frac{e^{-ct}}{(a-c)(b-c)}$ | $\frac{1}{(s+a)(s+b)(s+c)}$ |
| 12 | $\sin\omega t$, a sine wave | $\frac{\omega}{s^2+\omega^2}$ |
| 13 | $\cos\omega t$, a cosine wave | $\frac{s}{s^2+\omega^2}$ |
| 14 | $e^{-at}\sin\omega t$, a damped sine wave | $\frac{\omega}{(s+a)^2+\omega^2}$ |
| 15 | $e^{-at}\cos\omega t$, a damped cosine wave | $\frac{s+a}{(s+a)^2+\omega^2}$ |
| 16 | $\frac{\omega}{\sqrt{1-\zeta^2}}e^{-\zeta\omega t}\sin\omega\sqrt{1-\zeta^2}\,t$ | $\frac{\omega^2}{s^2+2\zeta\omega s+\omega^2}$ |
| 17 | $1-\frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega t}\sin\left(\omega\sqrt{1-\zeta^2}\,t+\phi\right)$, $\cos\phi=\zeta$ | $\frac{\omega^2}{s(s^2+2\zeta\omega s+\omega^2)}$ |

2 If we have two separate $s$ terms then the corresponding time function is the sum of the time functions corresponding to each of the separate $s$ terms. For example, if we have $[1/(s+2)]+[1/(s+1)]$ then the time function is the time function for $[1/(s+2)]$ plus the time function for $[1/(s+1)]$.

**Example**

A system gives an output of $1/(s + 5)$. What is the output as a function of time?

The output is of the form given in Table 3.1 as item 4 with $a = 5$. Hence the time function is $e^{-5t}$ and thus describes an output which decays exponentially with time.

**Example**

A system gives an output of $10/[s(s + 5)]$. What is the output as a function of time?

The nearest form we have in Table 3.1 to the output is item 5 as $2 \times a/[s(s + a)]$ with $a = 5$. Thus the output, as a function of time, is $2(1 - e^{-5t})$.

**Example**

A system has a transfer function of $1/(s + 2)$. What will be its output as a function of time when it is subject to a step input of 1 V?

The step input has a Laplace transform of $(1/s)$. Thus:

$$\text{Output } (s) = G(s) \times \text{Input } (s)$$

$$= \frac{1}{s+2} \times \frac{1}{s} = \frac{1}{s(s+2)}$$

The nearest form we have in Table 3.1 to the output is item 5 as $\frac{1}{2} \times 2/[s(s + 2)]$. Thus the output, as a function of time, is $\frac{1}{2}(1 - e^{-5t})$ V.

**Example**

A system has a transfer function of $4/(s + 2)$. What will be its output as a function of time when subject to a ramp input of 2 V/s?

The ramp input has a Laplace transform of $(2/s^2)$. Thus:

$$\text{Output } (s) = G(s) \times \text{Input } (s)$$

$$= \frac{4}{s+2} \times \frac{2}{s^2} = \frac{8}{s^2(s+2)}$$

The nearest form we have in Table 3.1 to the output is item 7 when written as $4 \times 2/[s^2(s + 2)]$. Thus the output, as a function of time, is $4[t - (1 - e^{-2t})/2] = 4t - 2(1 - e^{-2t})$ V.

### 3.3.1 Partial fractions

A technique that is often required to put an $s$ function in terms which identify with forms, so enabling the corresponding time function to be

obtained in Table 3.1 is *partial fractions*. The term partial fractions is used for the process of converting an expression involving a complex fraction into a number of simpler fraction terms. This technique can be used provided the highest power of $s$ in the numerator of the expression is less than that in the denominator. When the highest power in the numerator is equal to or higher than that of the denominator, the denominator must be divided into the numerator until the result is the sum of terms with the remainder fractional term having a numerator with a lower power than the denominator.

There are basically three types of partial fractions:

1 The numerator is some function of $s$ and the denominator contains factors which are only of the form $(s + a)$, $(s + b)$, $(s + c)$, etc. and so is of the form:

$$\frac{F(s)}{(s+a)(s+b)(s+c)}$$

and has the partial fractions of

$$\frac{A}{(s+a)} + \frac{B}{(s+b)} + \frac{C}{(s+c)}$$

There is a partial fraction term for each bracketed term in the denominator. Thus, if we have $1/(s + a)(s + b)$ there will be two partial fraction terms.

2 There are repeated $(s + a)$ factors in the denominator and the expression is of the form:

$$\frac{F(s)}{(s+a)^n}$$

and has the partial fractions of:

$$\frac{A}{(s+a)^1} + \frac{B}{(s+a)^2} + \frac{C}{(s+a)^3} + \cdots + \frac{N}{(s+a)^n}$$

A multiple root expression has thus a partial fraction term for each power of the factor. Thus, if we have $1/(s + a)^2$ there will be two partial fraction terms; if we have $1/(s + a)^3$ there will be three partial fraction terms.

3 The denominator contains quadratic factors and the quadratic does not factorise, being of the form:

$$\frac{F(s)}{(as^2+bs+c)(s+d)}$$

and has the partial fractions of:

$$\frac{As+B}{as^2+bs+c} + \frac{C}{s+d}$$

The values of the constants $A$, $B$, $C$, etc. can be found by either making use of the fact that the equality between the expression and the partial fractions must be true for all values of $s$ and so considering particular values of $s$ that make calculations easy or that the coefficients of $s^n$ in the expression must equal those of $s^n$ in the partial fraction expansion.

**Example**

Determine the partial fractions of:

$$\frac{s+4}{(s+1)(s+2)}$$

The partial fractions are of the form:

$$\frac{A}{s+1}+\frac{B}{s+2}$$

Then, for the partial fraction expression to equal the original fraction, we must have:

$$\frac{s+4}{(s+1)(s+2)}=\frac{A(s+2)+B(s+1)}{(s+1)(s+2)}$$

and consequently:

$$s+4=A(s+2)+B(s+1)$$

This must be true for all values of $s$. The procedure is then to pick values of $s$ that will enable some of the terms involving constants to become zero and so enable other constants to be determined. Thus if we let $s = -2$ then we have

$$(-2)+4=A(-2+2)+B(-2+1)$$

and so $B = -2$. If we now let $s = -1$ then

$$(-1)+4=A(-1+2)+B(-1+1)$$

and so $A = 3$. Thus

$$\frac{s+4}{(s+1)(s+2)}=\frac{3}{s+1}-\frac{2}{s+2}$$

**Example**

Determine the partial fractions of:

$$\frac{3s+1}{(s+2)^3}$$

This has partial fractions of:

$$\frac{A}{(s+2)} + \frac{B}{(s+2)^2} + \frac{C}{(s+2)^3}$$

Then, for the partial fraction expression to equal the original fraction, we must have:

$$\frac{3s+1}{(s+2)^3} = \frac{A}{(s+2)} + \frac{B}{(s+2)^2} + \frac{C}{(s+2)^3}$$

and so consequently have:

$$3s + 1 = A(s + 2)^2 + B(s + 2) + C$$

$$= A(s^2 + 2s + 1) + B(s + 2) + C$$

Equating $s^2$ terms gives $0 = A$. Equating $s$ terms gives $3 = 2A + B$. and so $B = 3$. Equating the numeric terms gives $1 = A + 2B + C$ and so $C = -5$. Thus:

$$\frac{3s+1}{(s+2)^3} = \frac{3}{(s+2)^2} - \frac{5}{(s+2)^3}$$

**Example**

Determine the partial fractions of:

$$\frac{2s+1}{(s^2+s+1)(s+2)}$$

This will have partial fractions of:

$$\frac{As+B}{s^2+s+1} + \frac{C}{s+2}$$

Thus we must have:

$$\frac{2s+1}{(s^2+s+1)(s+2)} = \frac{As+B}{s^2+s+1} + \frac{C}{s+2}$$

and so:

$$2s + 1 = (As + B)(s + 2) + C(s^2 + s + 1)$$

With $s = -2$ then $-3 = 3C$ and so $C = -1$. Equating $s^2$ terms gives $0 = A + C$ and so $A = 1$. Equating $s$ terms gives $2 = 2A + B + C$ and so $B = 1$. As a check, equating numeric terms gives $1 = 2B + C$. Thus:

$$\frac{2s+1}{(s^2+s+1)(s+2)} = \frac{s+1}{s^2+s+1} - \frac{1}{s+2}$$

## 3.4 First order systems

A first order system has a differential equation of the form (see Section 2.3.3):

$$\tau\frac{dy}{dt} + y = kx$$

As a function of $s$ this can be written as:

$$\tau Y(s) + Y(s) = kX(s)$$

and so a transfer function of the form:

$$G(s) = \frac{Y(s)}{X(s)} = \frac{k}{\tau s + 1}$$

where $k$ is the *gain* of the system when there are steady-state conditions and $\tau$ is the *time constant* of the system.

When a first-order system is subject to a unit impulse input then $X(s) = 1$ and the output transform $Y(s)$ is:

$$Y(s) = G(s)X(s) = \frac{k}{\tau s + 1} \times 1 = k\frac{(1/\tau)}{s + 1/\tau}$$

Hence, since we have the transform in the form $1/(s + a)$, using item 4 in Table 3.1 gives:

$$x = k(1/\tau)\, e^{-t/\tau}$$

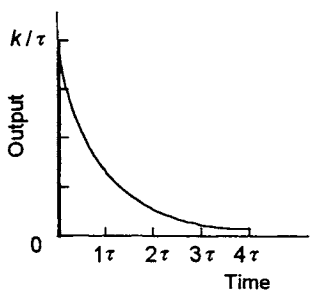


Figure 3.2 *Output with a unit impulse input to a first order system*

Figure 3.2 shows how the output varies with time; it is an exponential decay. The output rises to its maximum value at time $t = 0$ and then after $1\tau$ it drops to 0.37 of the initial value, after $2\tau$ it is 0.14 of the initial value and after $3\tau$ it is 0.05 of the initial value. Thus by about a time equal to four times the time constant the output is effectively zero. The exponential term tends to a zero value as the time $t$ tends to an infinite value.

When a first-order system is subject to a unit step input then $X(s) = 1/s$ and the output transform $Y(s)$ is:

$$X(s) = G(s)Y(s) = \frac{k}{s(\tau s + 1)} = k\frac{(1/\tau)}{s(s + 1/\tau)}$$

Hence, since we have the transform in the form $a/s(s + a)$, using item 5 in Table 3.1 gives:

$$x = k(1 - e^{-t/\tau})$$

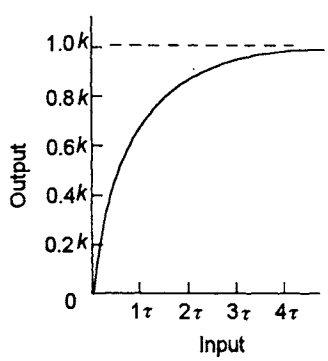


Figure 3.3 *Behaviour of a first order system when subject to a unit step input*

Figure 3.3 shows how the output varies with time. Initially, at time $t = 0$, the output is zero. It then rises to 0.63 of the steady state value after $1\tau$, then 0.86 of the steady state value after $2\tau$ and 0.95 of the steady state value after $3\tau$. After $4\tau$ the output is effectively at the steady state value

of $k$, the exponential term in the above equation becoming zero as time $t$ tends to infinity.

**Example**

A circuit has a resistance $R$ in series with a capacitance $C$. The differential equation relating the input $v$ and output $v_C$, i.e. the voltage across the capacitor, is:

$$v = RC\frac{dv_C}{dt} + v_C$$

Determine the output of the system when there is a 3 V impulse input.

As a function of $s$ the differential equation becomes:

$$V(s) = RCsV_C(s) + V_C(s)$$

Hence the transfer function is

$$G(s) = \frac{V_C(s)}{V(s)} = \frac{1}{RCs+1}$$

The output when there is 2 V impulse input is:

$$V_C(s) = G(s)V(s) = \frac{1}{RCs+1} \times 3 = \frac{3/RC}{s+1/RC}$$

Hence, since we have the transform in the form $1/(s + a)$, using item 4 in Table 3.1 gives:

$$x = (2/RC)\, e^{-t/RC}$$

**Example**

A thermocouple which has a transfer function linking its voltage output $V$ and temperature input of:

$$G(s) = \frac{30 \times 10^{-6}}{10s+1} \text{ V/°C}$$

Determine the response of the system when it is suddenly immersed in a water bath at 100°C.

The output as an $s$ function is:

$$V(s) = G(s) \times \text{input } (s)$$

The sudden immersion of the thermometer gives a step input of size 100°C and so the input as an $s$ function is $100/s$. Thus:

$$V(s) = \frac{30 \times 10^{-6}}{10s+1} \times \frac{100}{s} = \frac{30 \times 10^{-4}}{10s(s+0.1)} = 30 \times 10^{-4} \frac{0.1}{s(s+0.1}$$

The fraction element is of the form $a/s(s + a)$ and so the output as a function of time is:

$$V = 30 \times 10^{-4} (1 - e^{-0.1t}) \text{ V}$$

## 3.5 Second order systems

The differential equation for a second-order system is of the form:

$$\frac{d^2y}{dt^2} + 2\zeta\omega_n \frac{dy}{dt} + \omega_n^2 y = k\omega_n^2 x$$

where $\omega_n$ is the natural angular frequency with which the system oscillates and $\zeta$ is the damping ratio. Hence we have:

$$s^2Y(s) + 2\zeta\omega_n sY(s) + \omega_n^2 Y(s) = k\omega_n^2 X(s)$$

and so a transfer function of:

$$G(s) = \frac{Y(s)}{X(s)} = \frac{k\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

When a second-order system is subject to a unit step input, i.e. $X(s) = 1/s$, then the output transform is:

$$Y(s) = G(s)X(s) = \frac{k\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n)}$$

There are three different forms of answer to this equation for the way the output varies with time; these depending on the value of the damping constant and whether it gives an overdamped, critically damped or underdamped system (see Figure 2.13). We can determine the condition for these three forms of output by putting the equation in the form:

$$Y(s) = \frac{k\omega_n^2}{s(s+p_1)(s+p_2)}$$

where $p_1$ and $p_2$ are the roots of the quadratic term:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Hence, if we use the equation to determine the roots of a quadratic equation, we obtain:

$$p = \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2}$$

and so the two roots are given by:

$$p_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1} \quad \text{and} \quad p_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}$$

The important issue in determining the form of the roots is the value of the square root term.

1 $\zeta > 1$

With the damping factor $\zeta$ greater than 1 the square root term is real and will factorise. To find the inverse transform we can either use partial fractions to break the expression down into a number of simple fractions or use item 10 in Table 3.1. The output is thus:

$$y = \frac{k\omega_n^2}{p_1 p_2}\left[1 - \frac{p_2}{p_2 - p_1}e^{-p_1 t} + \frac{p_1}{p_2 - p_1}e^{-p_2 t}\right]$$

This describes an output which does not oscillate but dies away with time and thus the system is *overdamped*. As the time $t$ tends to infinity then the exponential terms tend to zero and the output becomes the steady value of $k\omega_n^2/(p_1 p_2)$. Since $p_1 p_2 = \omega_n^2$, the steady value is $k$.

2 $\zeta = 1$

With $\zeta = 1$ the square root term is zero and so $p_1 = p_2 = -\omega_n$; both roots are real and both the same. The output equation then becomes:

$$Y(s) = \frac{k\omega_n^2}{s(s + \omega_n)^2}$$

This equation can be expanded by means of partial fractions to give:

$$Y(s) = k\left[\frac{1}{s} - \frac{1}{s + \omega_n} - \frac{\omega_n}{(s + \omega_n)^2}\right]$$

Hence:

$$y = k[1 - e^{-\omega_n t} - \omega_n t e^{-\omega_n t}]$$

This is the critically damped condition and describes an output which does not oscillate but dies away with time. As the time $t$ tends to infinity then the exponential terms tend to zero and the output tends to the steady state value of $k$.

3 $\zeta < 1$

With $\zeta < 1$ the square root term does not have a real value. Using item 17 in Table 3.1 then gives:

$$y = k\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}}\sin\left(\omega_n\sqrt{(1 - \zeta^2)}\, t + \phi\right)\right]$$

where $\cos\phi = \zeta$. This is an under-damped oscillation. The angular frequency of the damped oscillation is:

$$\omega = \omega_n \sqrt{1-\zeta^2}$$

Only when the damping is very small does the angular frequency of the oscillation become nearly the natural angular frequency $\omega_n$. As the time $t$ tends to infinity then the exponential term tends to zero and so the output tends to the value $k$.

**Example**

What will be the state of damping of a system having the following transfer function and subject to a unit step input?

$$G(s) = \frac{1}{s^2 + 8s + 16}$$

The output $Y(s)$ from such a system is given by:

$$Y(s) = G(s)X(s)$$

For a unit step input $X(s) = 1/s$ and so the output is given by:

$$Y(s) = \frac{1}{s(s^2 + 8s + 16)} = \frac{1}{s(s+4)(s+4)}$$

The roots of $s^2 + 8s + 16$ are $p_1 = p_2 = -4$. Both the roots are real and the same and so the system is critically damped.

**Example**

A system has an output $y$ related to the input $x$ by the differential equation:

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = x$$

What will be the output from the system when it is subject to a unit step input? Initially both the output and input are zero.

We can write the Laplace transform of the equation as:

$$s^2Y(s) + 5sY(s) + 6Y(s) = X(s)$$

The transfer function is thus:

$$G(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^2 + 5s + 6}$$

For a unit step input the output is given by:

$$Y(s) = \frac{1}{s(s^2 + 5s + 6)} = \frac{1}{s(s+3)(s+2)}$$

Because the quadratic term has two real roots, the system is overdamped. We can directly use one of the standard forms given in Table 3.1 or partial fractions to first simplify the expression before using Table 3.1. Using partial fractions:

$$\frac{1}{s(s+3)(s+2)} = \frac{A}{s} + \frac{B}{s+3} + \frac{C}{s+2}$$

Thus, we have $1 = A(s + 3)(s + 2) + Bs(s + 2) + Cs(s + 3)$. When $s = 0$ then $1 = 6A$ and so $A = 1/6$. When $s = -3$ then $1 = 3B$ and so $B = 1/3$. When $s = -2$ then $1 = -2C$ and so $C = -1/2$. Hence we can write the output in the form:

$$Y(s) = \frac{1}{6s} + \frac{1}{3(s+3)} - \frac{1}{2(s+2)}$$

Hence, using Table 3.1 gives:

$$y = 0.17 + 0.33\,e^{-3t} - 0.5\,e^{-2t}$$

**Problems**

1 A system has an input of a voltage of 3 V which is suddenly applied by a switch being closed. What is the input as an $s$ function?

2 A system has an input of a voltage impulse of 2 V. What is the input as an $s$ function?

3 A system has an input of a voltage of a ramp voltage which increases at 5 V per second. What is the input as an $s$ function?

4 A system gives an output of $1/(s + 5)\ V(s)$. What is the output as a function of time?

5 A system has a transfer function of $5/(s + 3)$. What will be its output as a function of time when subject to (a) a unit step input of 1 V, (b) a unit impulse input of 1 V?

6 A system has a transfer function of $2/(s + 1)$. What will be its output as a function of time when subject to (a) a step input of 3 V, (b) an impulse input of 3 V?

7 A system has a transfer function of $1/(s + 2)$. What will be its output as a function of time when subject to (a) a step input of 4 V, (b) a ramp input unit impulse of 1 V/s?

8 Use partial fractions to simplify the following expressions:

(a) $\dfrac{s-6}{(s-1)(s-2)}$, (b) $\dfrac{s+5}{s^2+3s+2}$, (c) $\dfrac{2s-1}{(s+1)^2}$

9 A system has a transfer function of:

$$\frac{8(s+3)(s+8)}{(s+2)(s+4)}$$

What will be the output as a time function when it is subject to a unit step input? Hint: use partial fractions.

10 A system has a transfer function of:

$$G(s) = \frac{8(s+1)}{(s+2)^2}$$

What will be the output from the system when it is subject to a unit impulse input? Hint: use partial fractions.

11 What will be the state of damping of systems having the following transfer functions and subject to a unit step input?

(a) $\frac{1}{s^2+2s+1}$, (b) $\frac{1}{s^2+7s+12}$, (c) $\frac{1}{s^2+s+1}$

12 The input $x$ and output $y$ of a system are described by the differential equation:

$$\frac{dy}{dt} + 2y = x$$

Determine how the output will vary with time when there is an input which starts at zero time and then increases at the constant rate of 6 units/s. The initial output is zero.

13 The input $x$ and output $y$ of a system are described by the differential equation:

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = x$$

If initially the input and output are zero, what will be the output when there is a unit step input?

14 The input $x$ and output $y$ of a system are described by the differential equation:

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 3y = x$$

If initially the input and output are zero, what will be the output when there is a unit impulse input?

15 A control system has a forward path transfer function of $2/(s + 2)$ and a negative feedback loop with transfer function 4. What will be the response of the system to a unit step input?

16 A system has a transfer function of $100/(s^2 + s + 100)$. What will be its natural frequency $\omega_n$ and its damping ratio $\zeta$?

17 A system has a transfer function of $10/(s^2 + 4s + 9)$. Is the system under-damped, critically damped or over-damped?

18 A system has a transfer function of $3/(s^2 + 6s + 9)$. Is the system under-damped, critically damped or over-damped?

19 A system has a forward path transfer function of $10/(s + 3)$ and a negative feedback loop with transfer function 5. What is the time constant of the resulting first-order system?